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Hot Potato: An Analysis of the Conventional and Local Commodity Chains for Red Potatoes

Introduction

In the United States and around the world, few agricultural commodities have had the dominant and versatile presence in our diets as potatoes. Originating in South America, the potato has spread across the world and become the fourth highest produced crop (Djaman, K., & Koudah, K, 2021). Here in the United States, potatoes have been deeply incorporated into the food culture as seen by the significant versatility of the crop in our diets. Potatoes are boiled, baked, mashed, fried, and roasted and incorporated into popular dishes such as mashed potatoes, French fries, chips, tater tot hot dishes, casseroles, and more. But how are potatoes produced and how do they get to stores where we buy them? Are there differences between the potatoes available at the grocery store and those that are sold locally? To answer these questions, I examined a conventional and local commodity chain for red potatoes. For the conventional commodity chain, I investigated red potatoes sold by Green Giant because the company is recognizable throughout the country. For the local commodity chain, I chose to look at the red potatoes sold at the farmers markets by Schwartz Family Farms as they are a local producer. Through the examination of the commodity chain for conventional supplier Green Giant's red potatoes and red potatoes sold at a local farmers market, it becomes clear that sourcing potatoes locally is preferable.

Methods

The first step in analyzing the commodity chain for red potatoes both conventionally and locally produced began with locating the specific products that I would be working with. For the conventionally produced potatoes, I found a bag of Green Giant Fresh Red Potatoes at the Hy-Vee grocery store in Eagan, Minnesota. Based on the information on the bag's label and the information that I got from the Hy-Vee manager, Kaci Miller, I was able to trace the potatoes to being grown by Campbell Farms in

Farms to the shelves of Hy-Vee. The next step was to conduct research about Campbell Farms and learn about production of the potatoes in the bag. This was done through a combination of phone calls made to Campbell Farms as well as research to elaborate on the information that I received from them. During these phone calls, I spoke with Garret Koble at Campbell Farms.

For the locally produced potatoes, I visited the St. Paul Farmers Market and was able to find a vendor that sold red potatoes, Schwartz Family Farms. Due to the structure of farmers markets, I was able to speak directly with Melissa Peterson, one of the producers of the red potatoes that were being sold by Schwartz Family Farms. I combined this information with additional research to develop the commodity chain for these locally produced potatoes.

Conventional Path Analysis

Green Giant is a nationally recognized fresh, frozen, and canned vegetable producer. Green Giant's success is due in no small part to its recognizable branding and infamous jingle that debuted in the 1960s: "Ho, ho, ho, Green Giant" (Green Giant). Among the variety of products, the company sells are five-pound bags of Green Giant Fresh Red Potatoes that can be found lining the shelves of Hy-Vee, a Midwestern chain grocer, in Eagan, Minnesota. According to the bag's label, these potatoes are sold by Potandon Produce and are produced in Grafton, North Dakota.

The story of the potatoes in this Green Giant bag begins at Campbell Farms in Grafton, North Dakota.

Campbell Farms is a potato producer in North Dakota and Minnesota with 4,000 acres in production (Lynch, 2021). In 2011, Potandon Produce, the exclusive supplier of Green Giant Potatoes, was made the sales agent for Campbell Farms ("Potandon Names Exclusive Sales Agent for Campbell Farms", 2011). Through their relationship with Potandon Produce, Campbell Farms sells a portion of their potatoes under the Green Giant label (Koble, personal communications, November 3, 2022). Potandon Produce is a trans-national corporation that supplies potatoes and onions (Potandon Produce). In the United States, it is

the largest supplier of fresh potatoes and as of 2011 it had over ten percent market share (Potandon Produce). Through its partnerships with potato producers such as Campbell Farms,

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Potandon Produce has over 35,000 acres of agricultural land in production (“Potandon Names Exclusive Sales Agent for Campbell Farms”, 2011).

Campbell Farms was founded in 1978 when brothers Bill and Tom Campbell purchased 140 acres of land in Grafton, North Dakota, to produce red potatoes (Campbell Farms). Grafton is in the northeastern corner of North Dakota in the Red River Valley region, a prime area for potato production (Campbell Farms).

The Red River is 545 miles long and travels northward along the state border of North Dakota and Minnesota before joining Lake Winnipeg in Southern Canada (“Red River Geography”, 2015). The Red River Valley is the 17,000 square mile region that surrounds the river and is characterized by fertile soils and flat topography which are desirable to produce a variety of crops including potatoes (“Red River Geography”, 2015) (Schwert). The soil fertility in the region results from the presence of Glacial Lake Agassiz during the Pleistocene Epoch. Cold temperatures during this period allowed glaciers to expand into what is today the Red River Valley region which created the Glacial Lake Agassiz when they contracted (“The Valley”, 2015) (Schwert). Over time, sediment accumulated at the bottom of the lake and stimulated the growth of aquatic vegetation (“The Valley”, 2015) (Schwert). As the vegetation decayed and the process was repeated, the bottom of the Glacial Lake Agassiz accumulated biomatter and nutrients that created fertile soil and dense tall grass cover when the lake drained around 9,200 years ago (“The Valley”, 2015) (Schwert). Additionally, the expansion of glaciers into the Red River Valley region during the Pleistocene Epoch flattened the terrain making it easier to farm due to decreased soil erosion and the reduced risk of unequal irrigation and fertilizer application (“The Valley”, 2015) (Schwert). Based on archaeological artifacts, it is estimated that the first Indigenous populations inhabited the Red River Valley around 9,000 years ago, around the time that the Glacial Lake Agassiz drained (“Pembina State Museum: History”). The fur trade brought the first European settlers to the Red River Valley in the mid eighteenth century (“Pembina State Museum: History”) (Carter, 1968). The Red River Colony was

established as a Scottish colony in 1811 that brought further settlement to the region (Carter 1968).

The fertile soil and environmental conditions that characterized the landscape in Grafton provided a desirable context for Campbell Farms to grow potatoes which resulted in many successful harvests and

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allowed Bill and Tom Campbell to grow their operations. Today, Campbell Farms owns thousands of acres of land and has expanded to a second farm in Big Lake, Minnesota (Campbell Farms). Campbell Farms owns and operates every stage of the potato production process from growing and harvesting to processing, cleaning, bagging, and shipping (Campbell Farms). Whereas the Big Lake farm focuses on seasonally supplying the Twin Cities with fresh potatoes, the Grafton farm serves as the headquarters of Campbell Farms and supplies red and yellow potatoes year-round to consumers and large suppliers such as Green Giant (Campbell Farms). Thus, it makes sense that the Green Giant Red Potatoes come from Campbell Farms' Grafton location.

The growing process of red potatoes in Grafton begins immediately after last year's harvest when oats are planted as a cover crop over the winter (G. Koble, personal communication, November 3, 2022). Oats are an ideal cover crop in North Dakota because they can survive the harsh winter environments (Sustainable Agriculture Research and Education [SARE], 2007) (Wick, et al., 2018). The benefits of using oats as a cover crop include averting the run-off of phosphorus-rich soil particles and soluble nitrate, preventing erosion, providing the soil with carbon which increases organic matter and microorganism activity, suppressing weeds, and fixing moisture in the soil (SARE, 2007) (Wick, et al., 2018). In addition to cover crops, Koble stated that Campbell Farms manages soil fertility through an annual system of crop rotation that includes soybeans, wheat, and corn (G. Koble, personal communication, November 3, 2022). According to the Campbell Farms website, this system of crop rotation relies on the relationship with surrounding landowners (Campbell Farms). By rotating crops, the soil will be more fertile and have a higher nutrient availability because planting the same crop repeatedly creates nutrient imbalances and alters the ecology of the environment (Lessard, et al., 2000) (Franzen, et

al., 2021). Healthier soil has many positive impacts on potato production including increased yields and less need for chemicals such as pesticides and fertilizers in the growing process (Lessard, et al., 2000) (Franzen, et al., 2021). Additionally, planting the same crop repeatedly in the same area of land allows herbivorous pest populations to build up over time which can have a detrimental impact on potato

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harvests (Alyokhin, 2009). Thus, crop rotation allows Campbell Farms to produce potatoes more effectively and sustainably.

While Campbell Farms utilizes ecological-informed techniques to promote soil health, Koble mentioned that chemical fertilizers are also applied at the beginning of the growing season (G. Koble, personal communication, November 3, 2022). The chemical makeup and quantity of fertilizer applied depends on the results returned from soil and petiole testing to determine the available nutrients in the land (Franzen, et al., 2021). Additionally, land is treated with varying levels of fertilizers with different chemical compositions rather than a blanket application (Campbell Farms). Campbell Farms did not give specific information about the type of fertilizer that they use but based on fertilizer recommendations for potato production in North Dakota published by Franzen, et al., (2021), I estimate that Campbell Farms applies 120 pounds of nitrogen, 50 pounds of phosphorous pentoxide, and 50 pounds of potassium oxide per acre. Using estimates from Farm Management Specialist Ron Haugen (2022), the per acre cost was \$0.94 for nitrogen, \$0.65 for phosphorus, and \$0.53 for potassium for the 2022 growing season. Thus, the cost per acre for fertilizer was \$171.80 based on Campbell Farms' estimated needs, and the total cost for all its land under production is \$687,200. In addition to financial costs, chemical fertilizers have environmental consequences (Chandini, et al., 2019) (Nayana, S. & Ritu, S., 2017). Excess nitrogen can contaminate surface and groundwater and have health impacts such as cancer if ingested in large quantities (Chandini, et al., 2019) (Nayana, S. & Ritu, S., 2017). Additionally, interactions between soil bacteria and excess nitrogen can lead to emissions of carbon dioxide and the acidification of soil leading to reduced soil health, crop yields, and ecosystem damage (Chandini, et al., 2019) (Nayana, S. & Ritu, S., 2017). Phosphorous applications can cause eutrophication in water systems damaging aquatic ecosystems

(Chandini, et al., 2019) (Nayana, S. & Ritu, S., 2017).

After the soil management techniques are performed and the land is prepared to provide an ideal growth environment for potatoes, planting takes place during April and May (G. Koble, personal communication, November 3, 2022). Potatoes are often grown from seed potatoes which are tubers (the structure that stores the plant's nutrients becomes the potato) from last year's harvest that are reserved to

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be planted (Berzok, 2003). Each potato plant produces several tubers making this process possible (Berzok, 2003). Seed potatoes are planted with the leaf buds facing upwards (Berzok, 2003). According to Koble, Campbell Farms purchases their seed potatoes from Potandon Produce (G. Koble, personal communication, November 3, 2022). Seed potatoes often undergo multiple years of cultivation that include being grown in a greenhouse and in the field in order to multiply quality-controlled tubers (R. Johnk, personal communication, November 28, 2022). Seed potatoes are tested for quality and safety by state governments (R. Johnk, personal communication, November 28, 2022). Thus, there is a system of growing locations that the seed potatoes produced by Potandon Produce must travel through before they are shipped to Campbell Farms. The shipment of seed potatoes adds external input costs and fuel requirements to the production of potatoes. Koble did not mention where specifically the seed potatoes from Potandon come from, so I am unable to estimate the costs associated with the transportation of these seed potatoes.

The growing season for potatoes lasts approximately 20 weeks (Berzok, 2003). I was not able to get information from Campbell Farms on their irrigation practices. However, the precipitation levels during the growing season in the Red River Valley lead many potato producers to not use any external irrigation ("The Valley"). According to Koble, Campbell Farms uses insecticides to manage pests that impede the growth process (G. Koble, personal communication, November 3, 2022). The biggest threat to potatoes during the growing process in the United States is the Colorado Potato Beetle (Alyokhin, 2009) (Kadoić Balaško, M., et al., 2020). The Colorado Potato Beetle is native to Mexico, and the first outbreaks in United States potato fields occurred in the mid to late nineteenth century (Alyokhin, 2009) (Kadoić

Balaško, M., et al., 2020). The Colorado Potato Beetle spread quickly throughout the country and later to the rest of the world with increased trade and travel (Alyokhin, 2009) (Kadoić Balaško, M., et al., 2020) (Lessard, S., et al., 2000). The Colorado Potato Beetle feeds on the potato plant leaves, stems, and tubers which causes defoliation and destroys entire crops (Alyokhin, 2009) (Kadoić Balaško, M., et al., 2020) (Lessard, S., et al., 2000). Koble did not provide information about the specific chemicals used to manage the threat of potato pests on the harvest, but pesticides in general have detrimental environmental

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and health impacts (Nayana S. & Ritu, S., 2017). The consequences of pesticide use include killing non targeted plants and animals, crop and water contamination, soil degradation, and poisoning of those working directly with the chemicals (Nayana S. & Ritu, S., 2017). Pesticide usage can result in lowering effectiveness as well as pests develop resistance, requiring an increasing volume of chemicals to have the same effect, exacerbating environmental consequences (Nayana S. & Ritu, S., 2017).

Campbell Farms harvests their potatoes during the months of September and October which is consistent with other potato producers in the Red River Valley (“The Valley”) (G. Koble, personal communication, November 3, 2022). The first step to harvest the potatoes is to kill the plant in a process known as vine killing (Dittmar, P., et al., 2016) (Johnson, S. and Thornton, R. 2010). The potatoes are then left in the ground for an additional three weeks (Dittmar, P., et al., 2016) (Johnson, S. and Thornton, R. 2010). Vine killing reduces skin breaks and disease incidence, controls the size of tubers, and makes harvesting easier (Dittmar, P., et al., 2016) (Johnson, S. and Thornton, R. 2010). Campbell Farms did not provide specific information on their vine killing process, but the main methods are flail mowing and rolling, desiccation through herbicide applications, or a combination of the two (Dittmar, P., et al., 2016). From a food safety and environmental perspective, the first method is more desirable as it reduces chemical inputs. Then, the potatoes are dug up from the ground (Dittmar, P., et al., 2016). For large scale potato production as occurs at Campbell Farms, mechanical harvesters are used that can simultaneously dig up two to four rows of potatoes (Johnson, S. and Thornton, R. 2010). Mechanical harvesters are attached to the back of tractors and are driven through the fields (Johnson, S. and Thornton, R. 2010).

Once the potatoes are dug up, the mechanical harvester uses a system of conveyor belts to dump the tubers into trucks to transport the potatoes (Johnson, S. and Thornton, R. 2010). The mechanization allows Campbell Farms to efficiently harvest potatoes from their thousands of acres under production, but it results in greater equipment and energy costs. Related to the energy costs is the environmental impact that the mechanized harvesting process incurs due to its use of fossil fuels.

Once harvested, the potatoes are treated with a sprout suppressant according to Koble (G. Koble, personal communication, November 3, 2022). The most common sprout suppressant used on potatoes in

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the United States is chemical Isopropyl N-(3-chlorophenyl) carbamate (also known as chlorpropham) (United States Environmental Protection Agency [EPA], 1996). Chlorpropham suppresses the growth of sprouts on the potato after harvest (EPA, 1996). Chlorpropham is a slightly toxic compound according to the EPA (1996) and common symptoms of contact include eye and skin irritation. When ingested in high amounts, chlorpropham has shown symptoms such as incoordination, bloody tears, listlessness, and death in laboratory studies on mice (EPA, 1996). Chlorpropham bioaccumulates in water as it takes a long time to decompose and is moderately persistent in soil with high organic content (EPA, 1996). In the Red River Valley, the bioaccumulation of chlorpropham is significant because the region frequently floods (Schwert, D.). Thus, chlorpropham has the potential to build up in local water systems posing a significant health risk to people living in the area.

After the sprout suppressant treatment, the potatoes harvested at the Grafton location are moved to a storage facility located on site (G. Koble, personal communication, November 3, 2022). By providing a climate-controlled environment, these storage facilities allow Campbell Farms to supply consumers with potatoes year-round and not just during the harvest (Campbell Farms). For long term storage, potatoes need to be kept in a cool, dark environment to prevent disease and spoilage (Baghott, K.G., et al.). It should be mentioned, however, that warmer storage temperatures have benefits for certain time frames (Baghott, K.G., et al.). Keeping the potatoes at around sixty degrees Fahrenheit can heal damage caused during harvest and processing (Baghott, K.G., et al.). Thus, it could be desirable to incorporate a

period of wound healing in the storage process. Also, it is better to handle warm potatoes because tubers that are cool are more susceptible to damage (Baghott, K.G., et al.). Stored potatoes should therefore be warmed slightly before being handled (Baghott, K.G., et al.). It is also important to manage the amount of light that the potatoes are subject to because excess exposure can lead to greening (“Potato Greening Fact Sheet”). Direct sunlight can cause the buildup of chlorophyll on the exterior of the tuber which leads to a discoloration of the potato (“Potato Greening Fact Sheet”). This discoloration, called greening, turns off many customers from purchasing the potatoes which is for the better since this process also leads to the development of toxic alkaloids in the tuber (“Potato Greening Fact Sheet”).

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Once Campbell Farms is ready to supply the potatoes, the first step is to wash and package the stored potatoes. Campbell Farms washes their potatoes with water to remove excess dirt and other contaminants including chemicals located on the skin of the potatoes (G. Koble, personal communications, November 3, 2022). The potatoes are then dried and cooled to 45 degrees Fahrenheit (G. Koble, personal communications, November 3, 2022). To prevent the formation of condensation within the packages, a detrimental effect to the quality and safety of the potatoes, the packaging process takes place 24 hours after cooling (G. Koble, personal communications, November 3, 2022). The packaging process begins with the grading and sizing of the potatoes. Green Giant Red Potatoes are labeled as U.S. No. 1 Grade potatoes. According to the United States Department of Agriculture (USDA) (2011), U.S. No. 1 Grade potatoes have similar characteristics, and are firm, cleaned, and well-shaped. These potatoes are also not frozen and do not contain external or internal defects such as air cuts, blackheart, blight, bruising, internal discoloration, soft rot, and wet breakdown among others (USDA, 2011). Campbell Farms uses optical sorting techniques to detect visible flaws and X-rays to find deep imperfections (Campbell Farms). Green Giant red potatoes are at least one inch in diameter as well according to the bag. After sorting, the potatoes that meet the grade and size standards of Green Giant are then fed through computerized packaging machines (Campbell Farms). The packed bags are then cooled again 24 hours after the packaging process to further control for condensation (Campbell Farms).

Throughout the production process, Campbell Farms prides itself on its food safety measures. Campbell Farms is audited yearly by PrimusGFS for their compliance to the Global GAP, Global GMP, Food Safety Management Systems (FSMS), and Hazards Analysis and Critical Control Point (HACCP) programs (Campbell Farms). The Global GAP (Good Agricultural Practices) and FSMS programs seek to establish criteria for food safety and sustainability as well as the treatment of workers and animals (PrimusGFS) (“ISO 22000”). Similarly, Global GMP (Good Manufacturing Practices) focuses on assuring the quality of manufactured products (PrimusGFS) and the HACCP focuses on the control of hazards and contaminants to promote food safety (United States Food and Drug Administration [FDA]). These initiatives highlight Campbell Farms’ commitment to providing a safe, quality product to their

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customers. Campbell Farms also seeks to provide transparency in the growing process using the Produce Traceability Initiative (PTI) and providing significant information about the processes the potatoes undergo during production on their website (Campbell Farms). Employee safety is another value of Campbell Farms and there are several measures taken in this regard including, but not limited to, regular safety meetings and education and training regarding first aid, fires, hazards, and proper equipment use (PrimusGFS). Campbell Farms’ prioritization of food safety and quality, transparency, and employee safety highlights several positive aspects of this commodity chain and key differences with other conventional food producers that tend to obscure information.

Instead of sending potatoes to Potandon Produce, Campbell Farms ships orders of potatoes sold under the Green Giant label directly to buyers from their packing operations at the Grafton location, according to Koble (G. Koble, personal communication, November 3, 2022). A portion of the bags of Green Giant Fresh Red Potatoes are destined for the shelves of the Hy-Vee in Eagan, Minnesota. Hy-Vee is a chain grocer that owns and operates 285 supermarkets in the Midwestern states of Illinois, Iowa, Kansas, Minnesota, Missouri, Nebraska, South Dakota and Wisconsin (Hy-Vee). Hy-Vee generates over \$12 billion in sales annually and has around 93,000 employees (Hy-Vee) (“The Hy-Vee Story”).

The Green Giant Red Potatoes intended for the shelves of Hy-Vee in Eagan, Minnesota, are first transported in a tractor-trailer to the Hy-Vee Distribution Center in Chariton, Iowa (K. Miller, personal communication, November 10, 2022) (G. Koble, personal communication, November 10, 2022). The trip is roughly 650 miles and ten hours of driving time. Here, the potatoes are stored for up to three to four days before being sent to stores throughout the Midwest according to Hy-Vee produce manager Tom Crall (T. Crall, personal communication, November 15, 2022). Storage looks different at the distribution center because the potatoes are there for a short period of time and are handled frequently. Hy-Vee keeps the potatoes on the “warm side” according to Crall to protect them from damage during handling (T. Crall, personal communication, November 15, 2022). Additionally, Hy-Vee takes measures to cover the potatoes to block any incident light and prevent greening (T. Crall, personal communication, November 15, 2022). From the Hy-Vee Distribution Center, bags of Green Giant Red Potatoes are transported again

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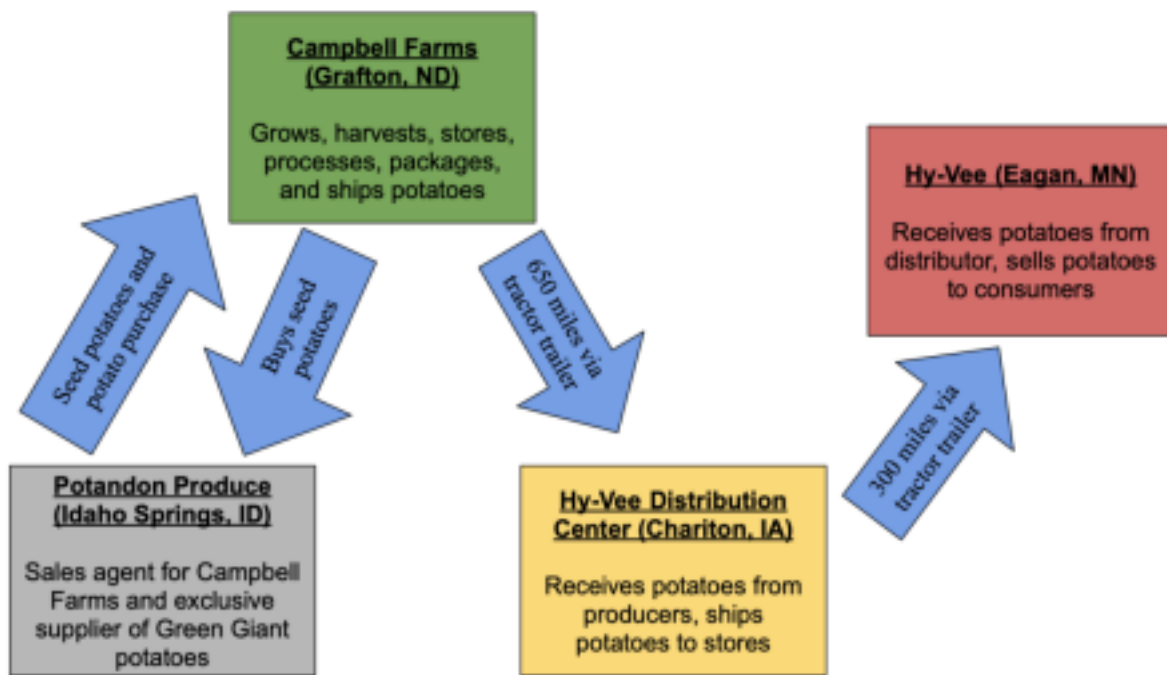
by tractor-trailers to grocery stores throughout the region (T. Crall, personal communication, November 15, 2022). Deliveries of Green Giant red potatoes are made two to three times a week to the Hy-Vee in Eagan, Minnesota according to store manager Kaci Miller (K. Miller, personal communication, November 10, 2022). This trip is roughly 300 miles and takes four and a half hours of driving time. In total, the Green Giant Red Potatoes travel 950 miles which takes approximately fourteen and a half hours of travel time.

Tractor-trailers, or semi-trucks, are transportation vehicles composed of a semi-trailer that is typically 53 feet long pulled by a tractor unit (“2012 Manac Trailer”). The significant size of the trailers is required to move the large amounts of products moved by corporations in the United States. For example, Crall estimates that Hy-Vee moves around a million pounds of potatoes per week (T. Crall, personal communication, November 15, 2022). Haugen (2022) estimated the average price of diesel in the Midwest to be \$2.90 per gallon for this year. Thus, the financial cost of diesel required in tractor-trailer transportation of Green Giant Red Potatoes from Grafton to Chariton to Eagan was \$423.40. It is important to note, however, that as these units can transport many bags of potatoes as seen in the size of

the trailer, the cost per bag is negligible. I was not able to get information from Campbell Farms about additional transportation costs, however Crall informed me that Hy-Vee pays freight fees for cargo incoming from suppliers and outgoing shipments to stores (T. Crall, personal communication, November 15, 2022). Tractor-trailer transportation has environmental costs as well. The consequences of the size of tractor-trailer are that these vehicles have poor fuel economy, getting around 6.5 miles per gallon of diesel (McElwain, C., 2022). Therefore, the trip that the potatoes took required 146 gallons of diesel fuel. The combustion of 146 gallons of diesel results in the emissions of 1.5 metric tons of carbon dioxide into the atmosphere according to the EPA (2022). Carbon emissions from diesel and other fossil fuel sources have led to climate change and damaged the atmosphere (EPA, 2022). Diesel usage causes ground-level ozone through the formation of oxides of nitrogen (NO_x) (EPA 2022). Ground-level ozone harms plants and animals and causes acid rain (EPA 2022). Exposure to diesel exhaust can also lead to respiratory conditions such as asthma (EPA 2022).

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Once the Green Giant Red Potatoes are received by the Hy-Vee in Eagan, Minnesota, they are stocked in the produce section. Consumers can buy a five-pound bag of these Green Giant red potatoes for \$3.99. According to the nutrition label, one medium potato weighing 148 grams has 110 calories. Thus, consumers that buy this bag of potatoes buy 422.47 calories per dollar. Additionally, each potato has around 45% of the recommended value for Vitamin C and has smaller amounts of iron, dietary fiber, calcium, and protein. In addition to availability and access, the ability to utilize potatoes is important as well requiring proper equipment and knowledge to clean and prepare them. While consumers can buy these potatoes at Hy-Vee or any grocery store that sells Green Giant potatoes, Campbell Farms also donates to local food banks to provide their product to those unable to afford it, a positive social impact of this commodity chain (Campbell Farms).



Local Path Analysis

In the heart of St. Paul, the capital city of Minnesota, the St. Paul Farmers Market operates year round to provide fresh local produce to the Twin Cities. Among the vendors that sell their produce at the

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market is Schwartz Family Farms. Schwartz Family Farms is a family farm that sells a wide variety of vegetables including red potatoes. According to Melissa Peterson, who runs Schwartz Family Farms' stand at the St. Paul Farmers Market, they employ a varying amount of people based on seasonal and commercial needs (M. Peterson, personal communication, November 20, 2022). The path that these red potatoes take are much less complicated than their Green Giant counterparts and highlight the difference between local and national commodity chains.

Schwartz Family Farms red potatoes are grown at their 50-acre farm that is located 40 miles north of St. Paul in Harris, Minnesota (M. Peterson, personal communication, November 13, 2022). Harris is located in the St. Croix River Basin. According to the National Parks Service (NPS) (Juckem, P., 2007), the soil in Harris can be characterized as loamy. Loam is soil that is composed of sand, silt, and clay, and

its benefits for agriculture include nutrient and water retention (Crawford, D. 2009) (Juckem, P., 2007). The St. Croix River was formed by erosion resulting from glacial activity and prehistoric seas that covered the area (Crawford, D. 2009) (Juckem, P., 2007). As the St. Croix River Basin was once at the bottom of a sea like the Red River Valley, this explains the presence of loamy soil in Harris as silt, sand, clay, and biomatter accumulated creating rich and fertile soil (Crawford, D. 2009) (Juckem, P., 2007). Fur trading and the logging industry brought European settlers to the area during the eighteenth and nineteenth centuries displacing Indigenous populations as was done in the Red River Valley (Crawford, D. 2009) (Juckem, P., 2007).

Like Campbell Farms, the growing process here begins with the application of a chemical fertilizer (M. Peterson, personal communication, November 13, 2022). As discussed previously, a combination of nitrogen, phosphate pentoxide, and potassium oxide is recommended to provide the plants with sufficient nutrients to grow (Franzen, D., et al. 2021). However, the application of chemical fertilizers incurs environmental and financial costs as well (Nayana S. & Ritu, S., 2017). Schwartz Family Farms utilizes polyculture farming methods, so I was not able to use the information provided by North Dakota State University to estimate their chemical fertilizer usage. I would expect that chemical fertilizer application rates to be lower for Schwartz Family Farms because of the ecological benefits of

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polycultures, and the overall amount to be lower due to size differences between the farms. Additionally, the potatoes are treated with an insecticide to prevent the infestation of the Colorado Potato Beetle, and Peterson emphasized the significant threat that pests present to the success of the crop (M. Peterson, personal communication, November 13, 2022). After the early growing season, no additional chemicals are used (M. Peterson, personal communication, November 13, 2022). According to Peterson, the theory is that by the harvest, the presence of the chemicals on the potato tubers will be diluted as it breaks down and is washed away (M. Peterson, personal communication, November 13, 2022).

Unlike Campbell Farms, Schwartz Family Farms grows a variety of different vegetables in addition to red potatoes including, but not limited to, onions, radishes, asparagus, carrots, squash, pumpkins, tomatoes,

cauliflower, rhubarb, cucumbers, and other potato varieties (M. Peterson, personal communication, November 13, 2022). Growing crops in polycultures fosters ecological relationships which provides natural pest control and reduces the need for chemical pesticides (Cardinale et al., 2011) (Horrigan et al., 2002). Polycultures also enhance soil fertility and biodiversity which delivers higher quality nutrient content to the crops grown in these environments. (Cardinale et al., 2011) (Horrigan et al., 2002). Growing in a polyculture also allows Schwartz Family Farms to rotate their crops with greater variety than Campbell Farms. As mentioned earlier, crop rotation has many benefits for potato production including improved soil health, increased yields, decreased chemical applications, and pest management (Lessard, et al., 2000) (Franzen, et al., 2021). Schwartz Family Farms does not grow corn or soybeans, two crops that are grown in monocultures in the conventional food system (M. Peterson, personal communication, November 13, 2022). Agricultural policies, most notably subsidies for corn and soybeans, have incentivized the overproduction of these two crops which has resulted in such monocultures (Barth, 2019). The absence of corn and soybeans in the fields of Schwartz Family Farms highlights their commitment to polyculture production and producing a diverse harvest of vegetables.

The growing season for Schwartz Family Farms similarly begins in April and May and ends in September and October when the potatoes are harvested (M. Peterson, personal communication, November 13, 2022). Seed potatoes are again used as the growing mechanism (M. Peterson, personal

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communication, November 20, 2022). According to Peterson, Schwartz Family Farms gets their seed potatoes from Johnk Seed Potatoes from Barnesville, Minnesota (M. Peterson, personal communication, November 20, 2022).

Johnk Seed Potatoes is run by Randy Johnk and is located roughly 200 miles northwest of Harris. The farm has around 1568 acres of land in production (Anderson, J., et al. 2021). Seed potatoes are grown over multiple years in order to control for disease and ensure a high-quality product by multiplying the tubers that come from desirable seeds (R. Johnk, personal communication, November 28, 2022). After each year, the potatoes are harvested and stored over the winter before being cut into smaller pieces each

with a bud and then replanted during the next growing season (R. Johnk, personal communication, November 28, 2022). Seed potatoes are labelled based on the number of years they have been grown in a field. For example, tubers that are in their second year of being cultivated in a field are labeled second generation, or G2, seed potatoes (R. Johnk, personal communication, November 28, 2022).

Johnk Seed Potatoes grow and sell G3 and G4 seed potatoes (R. Johnk, personal communication, November 28, 2022). Before Johnk receives their tubers, they have been grown in fields for two years at a remote farm in Syre, Minnesota, a small town 45 miles north of Barnesville. According to Johnk, this farm is far enough removed from any other potato production which lowers the disease risk as testing almost never reveals pathogens developing on the crop (R. Johnk, personal communication, November 28, 2022). But the seed potatoes go even farther back to a greenhouse in Halstad, Minnesota, a town located on the Minnesota-North Dakota border 37 miles northwest of Syre (R. Johnk, personal communication, November 28, 2022). In this greenhouse, mini tubers are grown in a controlled environment in the pre-nuclear stage (R. Johnk, personal communication, November 28, 2022). Once the mini tubers are ready to transition to the field, Johnk buys small bags of them, around 75 pounds, and sends them to the remote farm in Syre for the first two years of production (R. Johnk, personal communication, November 28, 2022). Then, the farm in Syre sends Johnk the G2 seed potatoes (R. Johnk, personal communication, November 28, 2022). The intermediary transportation of these seed potatoes totals 82 miles and one and a half hours of drive time.

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In terms of the production process, Johnk Seed Potatoes use conventional methods that include chemical fertilizers and pesticides (R. Johnk, personal communication, November 28, 2022). As previously stated, the use of these chemical inputs can bolster crop success but come at environmental and financial costs. After the growing season ends, Johnk harvests the seed potatoes and stores them in warehouses located on site in a cool, humid environment of around 38 degrees Fahrenheit and 90% humidity using a system of fans. (R. Johnk, personal communication, November 28, 2022). Johnk Seed Potatoes produces a select number of G4 seed potatoes, but typically only grows tubers for a year and

then sells the G3 seed potatoes (R. Johnk, personal communication, November 28, 2022). Johnk Seed Potatoes sells to many small farms east of the Mississippi River in the Midwest and southeastern United States (R. Johnk, personal communication, November 28, 2022).

During each stage of the production of seed potatoes, inspections are conducted by the Minnesota Department of Agriculture (R. Johnk, personal communication, November 28, 2022). According to Johnk, each state has a similar system of seed potato inspection run by the state government (R. Johnk, personal communication, November 28, 2022). In Minnesota, fields of seed potatoes are inspected for disease multiple times during the growing season (R. Johnk, personal communication, November 28, 2022). Additionally, a sample of the tuber stock is transported over the winter to Hawaii where they are grown and inspected for pathogens (R. Johnk, personal communication, November 28, 2022). These inspections ensure that safe seed potatoes are being sold to potato producers thus ensuring a quality product.

Johnk Seed Potatoes then send the seed potatoes 200 miles to Schwartz Family Farms in Harris. Here, the seed potatoes are planted and cultivated to produce consumable tubers. During the growing season, potatoes are irrigated, and no additional chemicals are applied after the initial applications (M. Peterson, personal communication, November 20, 2022). Irrigating the potatoes raises the input costs for machinery, water, and energy. However, these costs are small considering the scale of the farm. To harvest the potatoes, machines are used rather than manual labor (M. Peterson, personal communication, November 13, 2022). As discussed earlier, potato harvesting machines are pulled through fields by a

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tractor and dig up potatoes from the ground, moving them to a transportation unit (Johnson, S. & Thornton, R, 2010). Schwartz Family Farms likely has a smaller harvesting machine than Campbell Farms because they have less acres in production and produce a variety of crops on that land. The costs associated with mechanized harvesting include the price of machinery and fuel inputs to operate it. Its fuel requirements incur greater environmental costs than manual harvesting methods but are more time efficient for a small-scale operation that needs to focus on a wide variety of crops and products.

After the potatoes are harvested, Peterson says that Schwartz Family Farms stores them without cleaning just above freezing temperatures (M. Peterson, personal communication, November 20, 2022). This is consistent with potato storage recommendations as keeping potatoes at cold temperatures prevents the spread of disease (Baghott, K.G., et al.). Schwartz Family Farms uses a system of fans to keep the potatoes at the desired temperature (M. Peterson, personal communication, November 20, 2022). Cooling the potatoes is the only processing or storage measure taken by the farm after the potatoes are harvested (M. Peterson, personal communication, November 20, 2022).

Once Schwartz decides to sell the potatoes, they wash and clean the tubers (M. Peterson, personal communication, November 20, 2022). None of the harvested produce is sold to grocery stores, and instead Schwartz Family Farms sells at farmers markets throughout southeastern Minnesota (M. Peterson, personal communication, November 13, 2022). In addition to the St. Paul Farmers Market, Schwartz sells their produce at the Maple Grove, Woodbury, Eagan, Andover, Oakdale, Shoreview, Excelsior, White Bear Lake, Prior Lake, Chanhassen, Bloomington, and Savage farmers markets (M. Peterson, personal communication, November 13, 2022). The potatoes destined for the St. Paul Farmers Market are placed into box trucks that transport the produce (M. Peterson, personal communication, November 13, 2022). The potatoes are not packaged as they were at Green Giant. Packaging protects the potatoes from harm, but also creates new costs and waste. The box truck I saw in use was smaller than tractor-trailers which makes sense given the difference in size between Schwartz Family Farms and the corporations involved in the commodity chain for Green Giant. Although box trucks have poor fuel efficiency when compared with standard cars, their smaller size when compared to tractor-trailers means

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that they have higher fuel efficiency (“How to Save Gas While Driving Your Moving Truck”). The trip is roughly 40 miles and 45 minutes of driving time. Based on comparisons with the fuel efficiency of U Haul trucks, I estimate that box trucks have a fuel efficiency of about 10 miles per gallon of diesel (“How to Save Gas While Driving Your Moving Truck”). Thus, the transportation costs of this trip based on the earlier estimate for diesel prices are \$11.60 and 4 gallons of diesel. Again, the size of the box truck allows

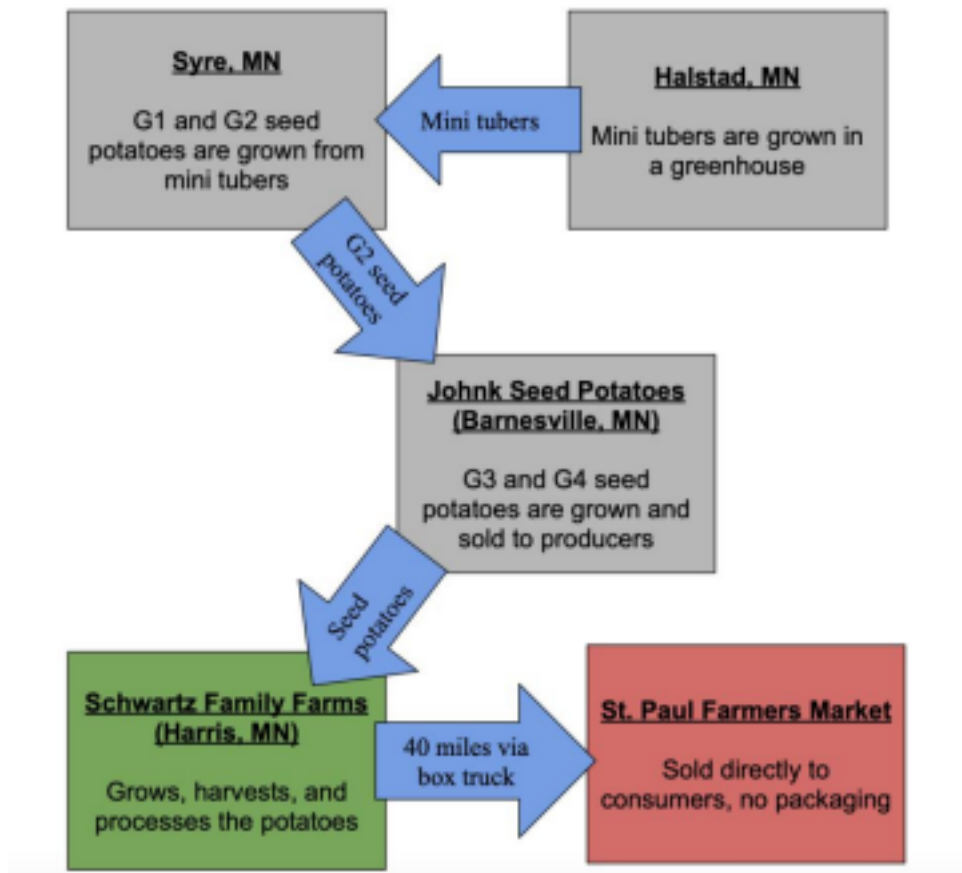
Schwartz Family Farms to transport enough produce to make the diesel costs negligible. Additionally, there are no extra financial costs for transportation as they drive the products to the St. Paul Farmers Market themselves, so freight fees do not exist. The environmental costs such as ground-level ozone and carbon emissions still apply in this situation, but to a much smaller degree as the trip only emits 0.04 metric tons of carbon dioxide (EPA, 2022).

At the St. Paul Farmers Markets, consumers can buy a group of red potatoes for \$4. The number of red potatoes appeared to be similar, though individual sizes varied. Nor was there a label that provides customers with information about the weight and nutritional profile to determine the value of the product. Based on information from the USDA (2019), the potatoes in the pack that I purchased would be classified as medium red potatoes and weighing an average of 213 grams. As there were ten potatoes in the group that I purchased, the unit cost of the product turns out to be \$0.85 per pound of red potatoes (\$4 for 4.7 pounds). The USDA (2019) also stated that each medium red potato has 150 calories, so each dollar spent for this product returns 375 calories.

Cost is not everything, however, and I believe that customer service and engaging with the community is very important for suppliers. When I went to the St. Paul Farmers Market to find a red potato vendor and buy some potatoes, I was a dollar short. Peterson immediately accepted the money that I did have and gave me the full group of red potatoes regardless. I still feel a tinge of guilt for accepting the offer as I feel that I shorted Schwartz Family Farms, but that one act made me as a consumer feel more connected with a vendor than anything that has ever happened to me while in a supermarket such as Hy-Vee. In addition, by going to a farmers market and interacting with the vendors, consumers are able to connect with the food production process (Falknor, 2021, pp. 13-19). The food system is more transparent

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when consumers can talk with producers, allowing people to make more informed decisions about what they eat and thus have greater influence in the industry (Falknor, 2021, pp. 13-19). Additionally, the St. Paul Farmers Market accepts SNAP benefits which creates greater food security as it provides access to fresh produce with good nutritional value.



Product Comparison

Despite being the same agricultural commodity, these red potatoes underwent significantly different processes. First, and probably most obvious, the two products were sold in different settings. The Green Giant Red Potatoes were sold in a regional supermarket chain in the suburbs of the Twin Cities whereas Schwartz Family Farms sold theirs at an outdoor market in downtown St. Paul. The consequences of selling at a farmers market as opposed to a supermarket chain include reduced transportation costs and energy use as products do not need to be shipped to intermediary distribution

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plants. It took 146 gallons of diesel to transport Green Giant Red Potato bags from the farm they were grown at to the shelves of Hy-Vee whereas only 4 gallons of diesel were needed for Schwartz Family Farms to move their produce. These reduced costs lead to a lower environmental impact. Schwartz Family Farms also only supplies to farmers markets in southeastern Minnesota which limits their transportation

and thus their environmental impact and financial costs when compared with Green Giant. Although the commodity chain for certain bags of Green Giant Red Potatoes is contained within the upper Midwest, both Campbell Farms and Potandon Produce transport products across the country (G. Koble, personal communications, November 3, 2022) (Potandon Produce). The frequent transportation for products outside of the studied commodity chain further exacerbates the differences between the two producers in terms of environmental impact. In addition, St. Paul Farmers Market is more accessible by public transportation than the Hy-Vee in Eagan, Minnesota which reduces the negative environmental impact of private transportation. Selling at farmers markets also allows for increased transparency in the food system as consumers can directly interact with the producers of their food (Falknor, 2021, pp. 13- 19).

Second, the level of processing and storage varied between the two products. Green Giant Red Potatoes underwent a more significant processing phase and were likely to be stored for a longer period when compared with those sold by Schwartz Family Farms. This is partially due to the greater transportation time of the Green Giant potatoes on their way from Grafton to Chariton to Eagan since the potatoes needed to be packaged to prevent the degradation of the product. In addition, Green Giant seeks to provide a year-round supply of potatoes, so they are potentially stored for several months at Campbell Farms prior to being shipped. Thus, processing measures including sprout suppressants such as chlorpropham were needed to be used by Green Giant to provide a quality product. Increased processing and storage requirements have several consequences. First, nutrient content in produce is highest once the potato is harvested and begins decreasing as it is served from the environment that provided it with these products. A higher storage time means potatoes have less nutritional value (McCurdy 2022). Second, the

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increased processing of the potatoes due the longer storage time has an adverse effect on freshness and safety of the potatoes (McCurdy 2022).

Third, the large, monoculture structure of Campbell Farms contrasts Schwartz Family Farms' small polyculture. Monoculture production allows Campbell Farms to produce a great volume of potatoes over the thousands of acres they have in production. On the other hand, polyculture production allows

Schwartz Family Farms to produce a wider variety of fresh produce and has ecological benefits including higher soil fertility and biodiversity and reduced chemical pesticide and fertilizer requirements (Cardinale et al., 2011) (Horrigan et al., 2002). In addition, as the farm is smaller, it allows Schwartz Family Farms to more effectively manage the land and the crops they grow as they have relatively more time to focus on each plant under cultivation.

The price and accessibility of red potatoes was another difference between the products. Hy-Vee has 285 locations located in eight states across the Midwest and they are only one of the suppliers of Green Giant Red Potatoes (“The Hy-Vee Story”). When compared to the prevalence of farmers markets across the country and their typical seasonality, it is clear that Green Giant Red Potatoes are more accessible than their local counterparts. Additionally, the labeling of the Green Giant Red Potatoes bag provides consumers with product information that is not as readily available for produce at farmers markets. The price of the two products is similar, but Green Giant Red Potatoes are slightly cheaper than their counterparts from Schwartz Family Farms as well, providing an estimated 50 calories more per dollar. However, the accessibility of information regarding the production process of potatoes and the steps it took from the farm was greater for Schwartz Family Farms and Johnk Seed Potatoes than it was for any of the large corporations involved in the conventional chain. As previously mentioned, one benefit of buying at the farmers market is being able to readily interact with the producers of your food and that is exactly what I did when I asked Peterson about the bag of red potatoes that I bought. Johnk Seed Potatoes was very open about their growing practices and the commodity chain as well. When I attempted to contact Campbell Farms, Potandon Produce, and Hy-Vee, I often faced answering machines, being put on hold and transferred, and less complete answers. This was in part due to the structure of these

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corporations. When I got into contact with these large corporations, I often was speaking to people without knowledge about their company’s practices or the broader commodity chain for potatoes because they were not involved in those processes. Thus, it was harder to obtain information from the conventional supply chain about the product they sold.

There are similarities as well between the red potatoes sold by Green Giant and Schwartz Family Farms. During the growing process, both Campbell Farms and Schwartz Family Farms use chemical fertilizers and insecticides to provide the potato plants with nutrients and protect it from pests such as the Colorado Potato Beetle. As previously stated, chemical fertilizers and insecticides can have negative health and environmental impact (Nayana S. & Ritu, S., 2017). On the positive side, both Campbell Farms and Schwartz Family Farms use some form of crop rotation in their production process. Another similarity is the contribution of both producers to greater food security within the community. Campbell Farms donates a portion of their potato harvest to local food banks and Schwartz Family Farms sells their produce at the St. Paul Farmers Markets which accepts SNAP benefits. Both produce potatoes in similar environments in the northern United States. Thus, similar environmental factors will affect the growing process for both farms.

Conclusion

The processes that red potatoes undergo from field to fork for conventional and local commodity chains bear some similarities but have striking differences. Perhaps the most significant are the differences in transportation, processing, and transparency between the two commodity chains that favor local production. However, the price and accessibility of these products should be noted as well which gives the advantage to the conventional supply chain. Based on these differences, I would recommend buying local red potatoes whenever possible, especially in areas where farmers markets are prevalent.

Examining the environmental impacts of the conventional supply chain, even one contained within the upper Midwest, with a local one highlights the vast differences in environmental consequences between the two methods. The lower processing of local red potatoes also leads to a fresher product that does not have the potential to be contaminated with harmful substances either. As consumers, our decisions have

massive repercussions on the sustainability of food systems, and whenever possible it is important to make environmentally conscious decisions. While it is important to note the cost and accessibility advantages than buying Green Giant Fresh Red Potatoes have, I believe they are negligible. First, the cost

for both bunch of red potatoes was \$4 and while I estimated to have gotten less from the farmers market, it wasn't a significantly lower amount. Factoring in taxes to the price of the Green Giant Fresh Red Potatoes that are not charged at the St. Paul Farmers Market makes it even more unclear which product was more cost effective. Second, depending on the location of the buyer, local potatoes may be more accessible than their Green Giant counterparts. For example, the St. Paul Farmers Market is easily accessible by public transportation and thus those that live in urban settings within the Twin Cities. In contrast, traveling to Hy-Vee often requires the use of a car as it is situated in the suburban environment of Eagan, Minnesota. Not only is this less accessible to people living in urban environments, but it also has additional environmental impacts as private transportation leads to significant carbon emissions. Additionally, the St. Paul Farmers Market accepts SNAP benefits making local potatoes accessible to people struggling financially. The commodity chain for the locally sourced red potatoes is not perfect as it still uses chemical inputs and fossil fuels for transportation and mechanized farming techniques, but the extent of these practices is dwarfed by the conventional supply chain. Just look at carbon emissions as a result of transportation fuel alone. Schwartz Family Farms emitted 97.3% less carbon dioxide in transporting their potatoes than that of the Green Giant potatoes. Making locally produced products such as potatoes more accessible by supporting local farmers and promoting the expansion of settings like farmers markets that allows the sale of these commodities should be a priority because it is clear that they support the health of the environment and the people.

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